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Advances in large amplitude oscillatory dilatational surface rheology – A review

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ABSTRACT

Large amplitude oscillatory dilatational (LAOD) surface rheology is an emerging tool for characterising the nonlinear mechanical behaviour of air-liquid and liquid-liquid interfaces in multiphase systems, such as in foams and emulsions. These interfaces, stabilised by surfactants or emulsifiers, exhibit complex viscoelastic properties that significantly influence bubble and droplet stability under large deformation conditions. While conventional methods mainly focus on linear viscoelastic behaviour, LAOD allows us to explore the nonlinear deformation behaviour of interfaces, offering deeper insights into the mechanical properties of interfaces upon large deformation.

This review highlights key advances in the dilatational rheology of interfacial films, providing a comprehensive background on LAOD experimental design, data collection, and analysis techniques. Recent developments in data analysis have revolutionised the LAOD technique. While Lissajous plots gave qualitative insights in the past, the new general stress decomposition (GSD) method allows for more quantitative analysis of the nonlinearities. GSD can quantitatively separate density-driven and actual rheological (i.e. network) contributions in the stress response. This separation reveals previously hidden rheological responses, allowing more accurate quantification of interfacial mechanics and deeper mechanistic insights into interfacial stabilisation phenomena. We will discuss how these LAOD methodologies have significantly enhanced our understanding of interfaces stabilised by surfactants, proteins, particles, and mixed systems.

The main take-home-message: The emergence of advanced LAOD methodologies has redefined how we characterise nonlinear interfacial mechanics. Future research should focus on expanding GSD applications and revisiting previously studied interfaces with GSD, which will quantify the previously qualitatively described behaviour of interfacial films.

1. Introduction

Multiphase systems are ubiquitous in life, and examples are emulsions and foams in pharmaceuticals, cosmetics, and foods [1,2,3,4]. Multiphase systems also play a key role in the function of the human body, for instance, where cell bilayers play a key role in cell functionality [5]. As a result, there is widespread research focus on these multiphase systems, with a specific emphasis on the boundary between the immiscible phases: the interface.

The stabilisation of liquid-liquid or gas-liquid interfaces is key in many applications. To successfully stabilise these thermodynamically unstable systems, stabilising molecules are needed, such as emulsifiers

Abbreviations: AFM, Atomic force microscopy; A-W, Air-water (interface); BAM, Brewster angle microscopy; Bo, Bond number; CCD, Charge-coupled device; DT, Drop tensiometer; E_d', Dilatational elastic modulus; E_d'', Dilatational viscous modulus; GSD, General stress decomposition; LAOD, Large amplitude oscillatory dilatation; LAOS, Large amplitude oscillatory shear; LVE, Linear viscoelastic; NLVE, Nonlinear viscoelastic; O-W, Oil-water interface; PL, phospholipid; S, strain stiffening/softening ratio.

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or surfactants. The chosen interfacial stabiliser will determine the interfacial properties and will, therefore, determine to a great extent the stability of emulsions and foams during deformations that may occur during emulsion or foam processing [6]. Depending on the type of stabiliser, the interface can form interfacial layers with different types of rheological/mechanical properties, and examples are elastic or viscoelastic layers. The interfacial mechanics play a major role in the stability of droplets or bubbles during deformations. Other external factors such as the system's pH, ionic strength or the presence of other components might further influence the interfacial properties [7,8,9,6] Understanding how a certain emulsifier behaves at the interface, taking external factors into account is thus crucial for understanding and predicting the process, temperature and storage stability of any multiphase system as well as to predict the behaviour of an emulsifier in stabilising an interface.

The behaviour of these interfacial stabilisers at the interface can be studied using rheology to obtain insights into the mechanical properties of an interface [10,11,12], imaging/microscopy techniques to obtain visual insights into the structure [13,14] and computationally to gain insights into molecular interactions and arrangements [15]. In this review, we will mainly focus on the mechanical properties, which are studied by interfacial (or surface) rheology.

In interfacial/surface rheology, an interface is deformed in a controlled manner while carefully analysing the stress response. Two main modes are commonly used: surface shear rheology and surface dilatational rheology [16]. In both, the deformations are commonly applied in oscillatory mode, where the deformation amplitude and frequency of oscillations can be varied. The deformation conditions can be used to test the mechanical response in two regimes: the linear viscoelastic (LVE) or the nonlinear viscoelastic (NLVE) regime. The LVE regime is the region where the interfacial microstructure recovers at the same rate as the disruption induced by the deformation. The NLVE regime is the regime where the disruption is more pronounced than the recovery, resulting in a breakdown of the microstructure. These regimes can be found by performing amplitude (or strain) sweeps, where the deformation amplitude is gradually increased. By measuring the first harmonic-based storage and loss moduli, one would find moduli independent of amplitude in the LVE regime (often <5 % deformation amplitude), and typically a decrease (and occasionally an increase) of moduli for the NLVE regime [17,18,19]. The rheological deformations in the NLVE regime are considered large deformations, often called large amplitude oscillatory dilatation (LAOD) or shear (LAOS).

During these large deformations, nonlinearities in the stress output emerge, due to the breakdown of the interfacial microstructure [16]. These nonlinearities often complicate the accurate analysis of interfacial film behaviour, especially when using conventional analysis methods, such as calculating the first harmonic-based moduli, which are often incorporated in commercially available rheological instruments. This inaccurate analysis is unfortunate, as deformation in the NLVE regime is often necessary to assess the stability of a system exposed to large deformations that occur during processing in industrial settings, such as the stability of air bubbles and oil droplets under high shear. Nonlinearities in the stress response can be accurately analysed in surface shear rheology, using methods previously established for nonlinear bulk shear rheology [20]. A well-known example is using Lissajous-(Bowditch) curves to qualitatively analyse the raw stress response. These curves can be further quantitatively characterised by determining shear stiffening/thickening factors and dissipated energy values [21]. In addition, methods have been derived to also include the higher harmonics, thus the nonlinearities, into so-called multi-harmonic moduli [22]. While in (surface) shear rheology, nonlinear analysis has received considerable attention, this is not the case for interfacial dilatational rheology.

In the past years, LAOD has seen increasing use, but it is still mostly studied in a qualitative fashion using Lissajous plots. This analysis already gives more insights into the mechanical properties of an interfacial layer, compared to solely studying the first-harmonic-based surface dilatational moduli. This technique combined with microstructure (imaging) analysis, using, for instance, ellipsometry, synchrotron scattering techniques, and electron or atomic force microscopy on deposited films, has yielded new insights into interface stabilisation ([23,18,24,14]). In addition, a new method has recently been developed to quantitatively analyse the nonlinearities of dilatational rheological output, which could help in further evolving the field of LAOD [25].

In this review, we aim to summarise these recent advances in LAOD rheology analysis and quantification and provide key insights into the dilatational rheological properties of interfacial films. We will first give a background on dilatational rheology, including analysis techniques, experimental design, and data analysis, to describe the state-of-the-art before diving into the recent developments. The latter will be extensively discussed to guide readers through the key elements of LAOD analysis in both qualitative and quantitative ways. Finally, we show how to relate the rheological findings to the microstructure of various types of interface stabilisers, such as small surfactants and large biopolymers (e.g., proteins and polysaccharides).

2. Introduction to large amplitude oscillatory dilatation (LAOD)

2.1. Dilatational surface rheology

Many destabilisation mechanisms in multiphase systems involve substantial changes in interfacial area. This is where dilatational interfacial rheology comes into play, which studies the mechanical properties of interfaces, such as liquid-liquid (e.g., oil-water) or gas-liquid (generally air-water) interfaces, under conditions where the total area of the interface is changed [26]. It focuses on the properties of the interfaces upon compression and expansion, when an all-sided in-plane compression or extension is performed. The surface mass is, therefore, not always conserved during these types of deformation [27]. Upon compression, the interfacial density of surface-active agents may increase, which can cause the desorption of some surfactants from the interface back into the bulk. Whether the molecules desorb back to the bulk or remain at the interface will be determined by the desorption energy. Analogously, upon extension, the interfacial density decreases, often facilitating the absorption of new molecules from the bulk to the newly created interfacial area.

Different instruments can be used to perform interfacial dilatational experiments [28]. The Langmuir-Blodget trough (commonly referred to as Langmuir trough) was the first instrument to be employed, based on the early experiments of Agnes Pockels on thin films of amphiphilic substances at the surface of water [29]. Irvin Langmuir and Katherine Blodgett then improved the equipment, which consists of a liquid reservoir, with movable barriers at the interface, that allow for the compression and expansion of the interface [30]. The dynamic surface tension (γ_d) during the deformations is measured using a Wilhelmy plate, connected to a balance [31]. However, the Langmuir trough does not provide a purely dilatational response, but a combination of shear and dilatation, due to the two barriers compressing the interface from just two sides, essentially performing a uniaxial compression rather than an all-sided one [32]. A fully all-sided deformation can be achieved by an equal deformation from all directions, which can, for instance, be achieved by using a recently invented quadrotrough [33].

A different method to measure surface tension was described in two early works by [34,35], based on the so-called "oil-drop" method, where the surface tension was for the first time calculated in real-time to measure lipase kinetics from the contour of a rising oil droplet in water, also known as an inverted pendant drop. Based on this rising drop setup, a dynamic drop tensiometer (DT) was later developed [36], where a droplet of a few millimetres in diameter at the tip of a needle could be deformed by changing the volume, using a syringe, thereby expanding or contracting the interface. These deformations are, in general, a better approximation for an all-sided extension/compression, than those

applied in the Langmuir trough, if they are applied affinely.

The basic setup of a dynamic DT consists of a cuvette, where the oil droplet is formed at the tip of a U-shaped syringe, and a high-speed or charge-coupled device (CCD) camera to record the oil drop shape within the water phase. Through the Young-Laplace equation, the surface tension of the droplet can be obtained:

$$P^2 - P^1 = 2\gamma_d H + \Delta\rho g h \quad (1)$$

where P^2 is the pressure inside the droplet, P^1 is the pressure outside the droplet, γ_d is the dynamic surface tension, H is the curvature of the interface, $\Delta\rho$ is the density difference between the two phases $(\rho_d - \rho)$, ρ_d is the density inside the droplet, ρ is the density of the outer phase, and h is the height difference [37,27,38].

So far, we have defined what interfacial dilatational rheology is, and the main instruments that have been developed to study this. In the next section, we explain *how* to perform dilatational rheology experiments, with a focus on using drop tensiometry.

2.2. Experimental design: Dilatational rheology using drop tensiometry

When working with a drop tensiometer, the measurement can be carried out using either the rising/emerging drop/bubble/inverted pendant drop method or the pendant/hanging drop method [26]. In the rising method, a droplet of the lower density phase or an air bubble is created at the tip of a curved needle, immersed in a cuvette filled with the higher density phase (often water). The phases are reversed when using the pendant drop method, where a droplet of the high-density phase is created at the tip of a straight needle, which is surrounded by the lower-density phase (which can be oil or air). The choice of method depends on several factors, such as 1) the available amount of emulsifier or oil phase, where the phase with a lower availability is typically put in the droplet, and 2) transparency (the continuous phase should be sufficiently transparent to capture the droplet contour).

The exact calibration of a drop tensiometer is vital for obtaining accurate measurements (Fig. 1). For the correct calculation of the interfacial tension, we require knowledge of the horizontal (cr) and vertical (cz) calibration factors for the conversion of the CCD-camera pixels to SI units (millimetres) [39]. These parameters are determined during the optical calibration. The optical calibration of the instrument

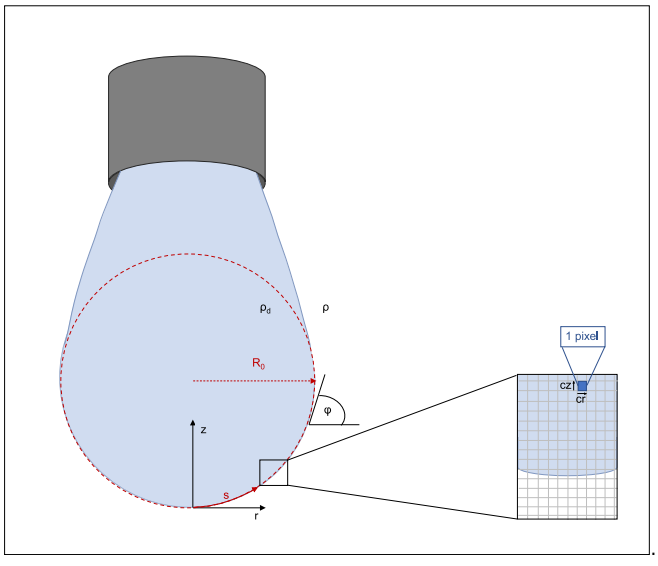


Fig. 1. Schematic overview of the surface tension determination in a drop tensiometer. A droplet is formed at the tip of a capillary in a surrounding liquid or gas phase. The figure demonstrates how the variables in the Young-Laplace equation are obtained. The zoomed in panel demonstrates the pixel analysis in modern drop tensiometry software.

can be done in three different ways: calibration with a sphere of known radius, calibration with a capillary of known radius, or calibration with a pure liquid or water at a specific temperature. As reported by [39], the accuracy of such conversion factors should be better than 0.1 %.

For the measurement, either a high-speed camera or a CCD camera is used to take pictures of the given droplet at fixed intervals. A corresponding software determines the coordinates of the drop profile by image analysis and sub-pixel edge detection, using cr and cz. The interfacial tension is then calculated from this information by fitting the profile to the Young-Laplace Equation, which relates the Laplace pressure across an interface to the curvature of the interface and the interfacial tension [40].

By taking advantage of axisymmetry, the Young-Laplace Equations can be expressed in terms of the cylindrical coordinates r and z (the radial and vertical coordinate, respectively), together with the tangent angle φ [41]. To facilitate numerical solutions and generalisation, these equations are typically written in dimensionless form by scaling all lengths with the radius of curvature at the apex R_0. The following set of coupled dimensionless differential equations then describes the drop profile as described by Berry et al. [41]:

$$\frac{d\varphi}{d\bar{S}} = 2 - Bo\bar{z} - \frac{sin\varphi}{\bar{r}} \quad (2)$$

$$\frac{d\bar{r}}{d\bar{S}} = cos\,\varphi \quad (3)$$

$$\frac{d\bar{z}}{d\bar{S}} = sin\,\varphi \quad (4)$$

where $\bar{r} = r/R_0$ is the dimensionless radial coordinate, $\bar{z} = z/R_0$ is the dimensionless vertical coordinate, and $\bar{S}/R_0$ is the dimensionless arc length measured from the drop apex. The Bond number (Bo) or Eötvos number, is defined as

$$Bo \equiv \frac{(\rho_d - \rho)gR_0}{\gamma_d} \quad (5)$$

where ρ_d is the density inside the droplet, ρ is the density of the outer phase, g is the gravitational acceleration, R_0 is the radius of curvature at the droplet's apex and γ_d is the interfacial tension. It is a measure of the relative importance of gravitational forces to interfacial forces. It can be used to evaluate the quality of the obtained data (and the experimental setup). A small Bond number implies that the interfacial forces dominate the gravitational forces; the interfaces are only slightly perturbed from a spherical shape, and the sensitivity of the method is reduced [41,42]. Acceptable bond numbers range from 0.1 to 1. (Berry et al., [41] reported, that the Bond number is strongly influenced by the needle width. With decreasing needle width, the minimum true Bond number required for accurate calculation also decreases.

By applying a sinusoidal (oscillatory) deformation of the volume or area of the created drop, the resulting sinusoidal response of the surface tension can be used to calculate a complex surface dilatational modulus, E_d^*, defined as

$$E_d^* = \frac{\mathcal{F}(\Pi(t))}{\mathcal{F}(A^{-1}\Delta A(t))} = E_d' + iE_d'' \quad (6)$$

where $\mathcal{F}$ denotes a Fourier transform, $\Pi(t) = \gamma_d(t) - \gamma_{eq}, \gamma_{eq}$ is the surface tension of the nondeformed interface, A is the interfacial area, E_d' is the dilatational storage or elastic modulus (in phase with the oscillatory deformation) accounting for the dilatational elastic contribution, $E_d'' = \omega\eta_d$ is the dilatational loss or viscous modulus (90° out of phase with the oscillatory deformation), accounting for the dilatational viscous contribution, with ω being the frequency and η_d is the dilatational viscosity [32,43,44]. The dilatational modulus is the resistance of the interface against all-sided expansion/compression, and hence the two-

dimensional equivalent of the bulk modulus.

The volume and area can be set as parameters to control the oscillatory deformations. In the case of area-based oscillations, the equipment requires a feedback loop, where the area is constantly calculated from the drop image to make adjustments from the starting point i.e. the set area. In volume-based oscillations, regulation is not performed via image analysis but through communication between the motor and software, resulting in the controlled rotation of the motor shaft over time. For this reason, a volumetric calibration for the syringe, where the volume expelled by a syringe during one revolution of the motor is measured, is mandatory before starting the experiment. In other words, one has to measure the exact volume being pumped in/out by the motor before selecting a volume-based regulation for the experiment.

The downside of volume-based oscillations is that, as shown in the previous equation, the dilatational modulus is calculated based on area changes, and volume-based changes can result in less precise sinusoidal oscillations of the area. In particular, a drift from the set amplitude is possible between oscillation cycles i.e. the set value of the amplitude is not reached. On the other hand, a volume-based oscillation usually leads to a less noisy data set and allows for very small periods (as small as 1 % deformation amplitude), while small periods cannot be reached with an area-based oscillation. An area-controlled oscillation, however, minimises deviations from the setpoint value. In short, each regulation method comes with advantages and disadvantages, and one should carefully choose a regulation type based on the required experimental setup. The volume-based or area-based regulation can also be used for simple surface tension measurements without oscillation to keep the area/ volume constant over time, e.g. during film formation.

2.3. Linear and nonlinear dilatational deformations

Three types of dilatational rheology tests are commonly performed: time, frequency, and amplitude sweeps. A time sweep allows one to follow the evolution of interfacial stiffness over time by plotting E_d' and E_d'' as a function of time or surface pressure [45,46,19], where the latter is defined as $\Pi_0 = \gamma_0 - \gamma(t)$, and γ_0 is the surface tension of the clean interface. This sweep is measured by subjecting the droplet or bubble to small oscillations with a fixed frequency and amplitude (Fig. 2, type 2).

It is worth noting that some literature refers to the monitoring of the dynamic surface tension over time also as “time sweep” tests, where no oscillations are performed, and thus this should be referred to as “interfacial tension” or “surface tension” measurements instead (Fig. 2, Time sweep type 1).

The earlier-mentioned interfacial/surface tension analysis can be combined with rheological tests, such as a frequency sweep, amplitude sweep, or step-dilation. In the section below, we will focus on the oscillatory rheology via frequency and amplitude sweeps. For more information about step dilatation, we refer the reader to [47]. First, the interface would be allowed to age, with typical times ranging from 1 to 16 h, followed by a desired rheological test ([48,49,50,51,52]). Frequency sweeps are used to assess the dependence of the dilatational moduli of the interface on frequency ω. Generally, the frequency of the deformation is increased, while keeping the amplitude constant. The dependence of the complex dilatational modulus on frequency can reveal information on the nature of interfacial viscoelasticity. In the absence of in-plane interactions, surfactants follow the Lucassen-Van den Tempel model [53], where the complex modulus is dominated by bulk-interface exchange. This model introduces a characteristic frequency, ω_c, which marks the transition between diffusion-controlled dynamics at lower frequencies and other interfacial processes dominating at higher frequencies. Far below the characteristic frequency, the Lucassen-Van den Tempel model scales as $E_d' \sim \omega^{0.5}$. For solid-like interfaces, the frequency dependence of the E_d' often scales in a power-law manner $E_d' \sim \omega^n$, with $n \ll 0.5$. A low exponent n (~0.1–0.2) indicates the interface has the characteristics of a disordered solid (i.e., a 2D gel or jammed system), and is typically a sign that the interfacial structure displays a broad spectrum of relaxation times [54,55]. In such interfaces, relaxation is not typically governed by bulk-interface exchange, and other processes play a more dominant role in the interfacial elasticity, such as momentum transfer between interface and bulk or in-plane interactions [47]. For experiments using drop tensiometry, the maximum frequency that can be accurately applied is 0.1 Hz, whereas the lower limit employed is usually around 0.001 Hz, as lower frequencies would increase the measuring time, thus introducing the effects of further ageing of the interface or adsorption of additional surface-active molecules. Faster frequencies up to 150 Hz can be reached using bubble pressure analysis in combination with a piezo-controlled motor [56].

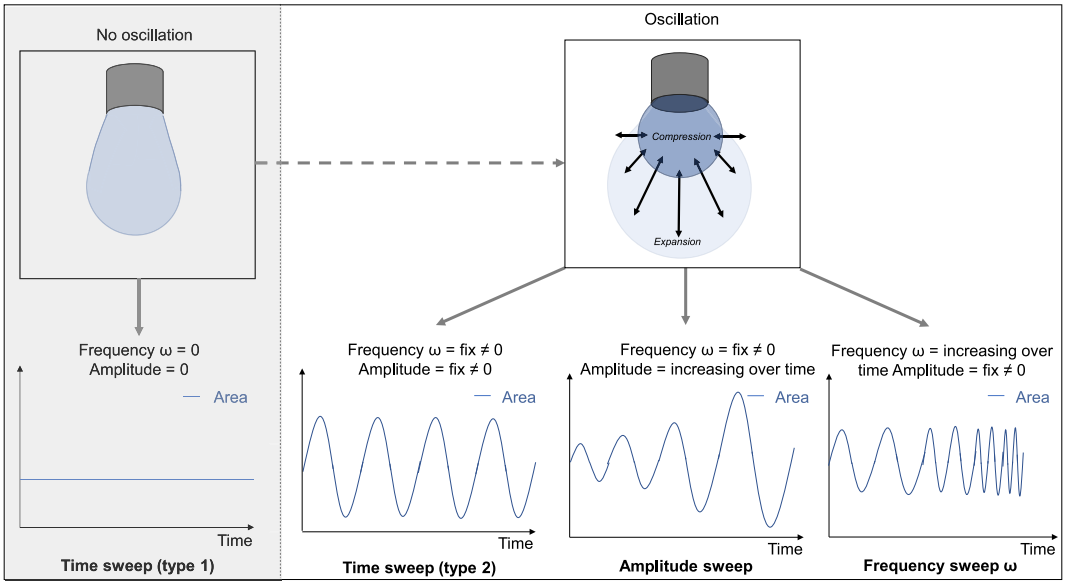


Fig. 2. Overview of the commonly performed rheology tests: time sweep, frequency & amplitude sweep. These tests are used to characterise the viscoelastic properties of interfacial layers. A time sweep monitors changes in the interfacial modulus over time at constant frequency and deformation amplitude, typically to observe adsorption or film formation. A frequency sweep probes the interface’s viscoelastic behaviour across a range of frequencies and a constant deformation amplitude. An amplitude sweep probes the interface’s viscoelastic behaviour across a range of deformation amplitudes and a constant frequency.

Amplitude sweeps, where ω is kept constant, and the amplitude of oscillation is gradually increased from small to large deformations, are commonly used to find the limit of linearity, thus defining the linear (LVE) and nonlinear viscoelastic (NLVE) regimes [57,58,45,18,50]. This method is mostly used to ensure that other tests, such as time and frequency sweeps, are performed within the linear regime. The interfacial viscoelasticity as a function of amplitude is determined by oscillating the droplet or air bubble with a fixed frequency (0.01 to 0.1 Hz) and varying amplitudes from usually 1–30 %. In amplitude sweeps, the applied frequency and deformation regime are relatively narrow to avoid the formation of surface waves [16]. In the LVE regime, it is assumed that the interfacial stress response amplitude is proportional to the applied deformation amplitude. The moduli are calculated from the intensity and phase of the first (or fundamental) harmonic in the frequency spectrum [16]. In this regard, the first harmonic is the input frequency of oscillation defined by the experimental settings.

Amplitude sweeps also provide a plethora of information when the deformations are performed in the NLVE regime, which goes beyond the quantification of E_d' and E_d'', which, as will be indicated below, are not accurate anymore. This corresponds to large amplitude oscillatory dilatation (LAOD). Interfaces show a nonlinear dynamic response beyond a critical amplitude as a consequence of the changes in the interfacial microstructure induced by the area changes. This nonlinear behaviour conveys additional information about the interfacial viscoelasticity and its stability under real process conditions.

In the NLVE regime, the surface stress response can be decomposed into a superposition of sinusoidal waves, starting with the applied frequency and including positive integer multiples of this frequency, known as higher harmonics. Using the Fourier Transformation, the contribution of such higher harmonics to the stress signal can be accurately quantified. Each of these higher harmonics has its own amplitude and phase angle [57]. Analysing highly nonlinear behaviour in dilatational experiments can pose challenges, as the applied deformation may become non-affine, as demonstrated for very stiff (hyper)elastic interfacial films [59]. The result is a non-uniform, thus non-Laplacian shape of the droplet, thereby hindering the accurate analysis of surface stress upon oscillatory deformations [60]. Examples of a Laplacian and non-Laplacian droplet shape are shown in Fig. 3. In such cases, capillary meniscus dynamometry offers better results. This method uses both the shape of the meniscus and the internal capillary pressure to calculate the components of the surface stress in the meridian or parallel directions [61].

When looking at the literature, for time sweeps and frequency sweeps, it is often tacitly assumed that the data was obtained within the LVE regime, and, therefore, that the surface tension signal follows a simple sine wave, which can be Fourier transformed to obtain E_d' and E_d'' from the amplitude and phase of the first harmonic [62]. If these tests are performed outside of the LVE regime, then E_d' and E_d'' become inaccurate, due to the presence of higher harmonics in the signal [16]. Unfortunately, the extent of the LVE is often not checked in studies that perform dilatational rheology, and frequency or time sweeps are sometimes performed at amplitudes as large as 20 %. These measurements might well be in the nonlinear regime, therefore making the obtained E_d' and E_d'' values questionable [16]. A combination of tests within a study is, therefore, the most recommended approach. (Rheological) time sweeps are used to evaluate a proper plateau for the ageing of the interface, ensuring the interface does not evolve to a large extent during a frequency or amplitude sweep, which would render the results inaccurate. Amplitude sweeps should then be used to evaluate the extent of the LVE regime and analyse the behaviour of the interface under large deformations. Finally, frequency sweeps can be used to study the time-dependent rheological behaviour of the interface at an amplitude in the LVE regime.

Since the dynamic moduli lose their physical meaning beyond the LVE, other approaches are necessary to evaluate the nonlinear deformation behaviour within the LAOD regime, which will be discussed in the next sections.

2.4. Drawbacks of using the first harmonic-based moduli in the NLVE

Within the LVE regime, the surface tension signal follows a simple sine wave, which can be Fourier-transformed to obtain E_d' and E_d'' [62]. Once the deformation is in the nonlinear regime (large amplitudes), the stress signal is no longer a simple sinusoidal response, but consists of contributions at the applied testing frequency and terms at multiples of this frequency [63,16] (Fig. 4). Some (modern) drop tensiometer

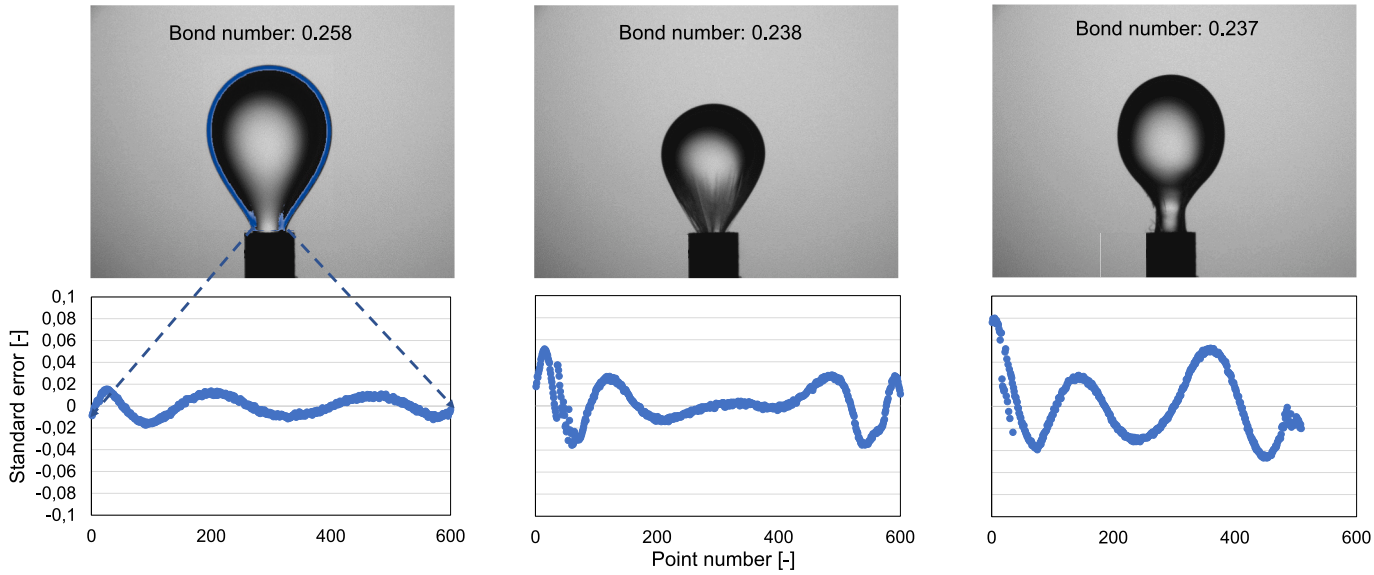


Fig. 3. Examples of a rising oil droplet (known as inverted pendant drop) in water, that have a Laplacian (left) and non-Laplacian shape (middle and right), captured in a drop tensiometer. The bottom rows show the "point number". To clarify, the droplet image was first processed in pixel coordinates to detect the droplet interface. Each pixel-derived contour point was then assigned to a consecutive index ('point number'), which serves solely as a positional identifier. This enables localisation of the fitting error from the Laplace equation along the contour and ensures that the points remain in the correct order when computing local curvature and fitting the theoretical shape. Importantly, the point number is not a fitting parameter and is independent of image resolution or pixel values.

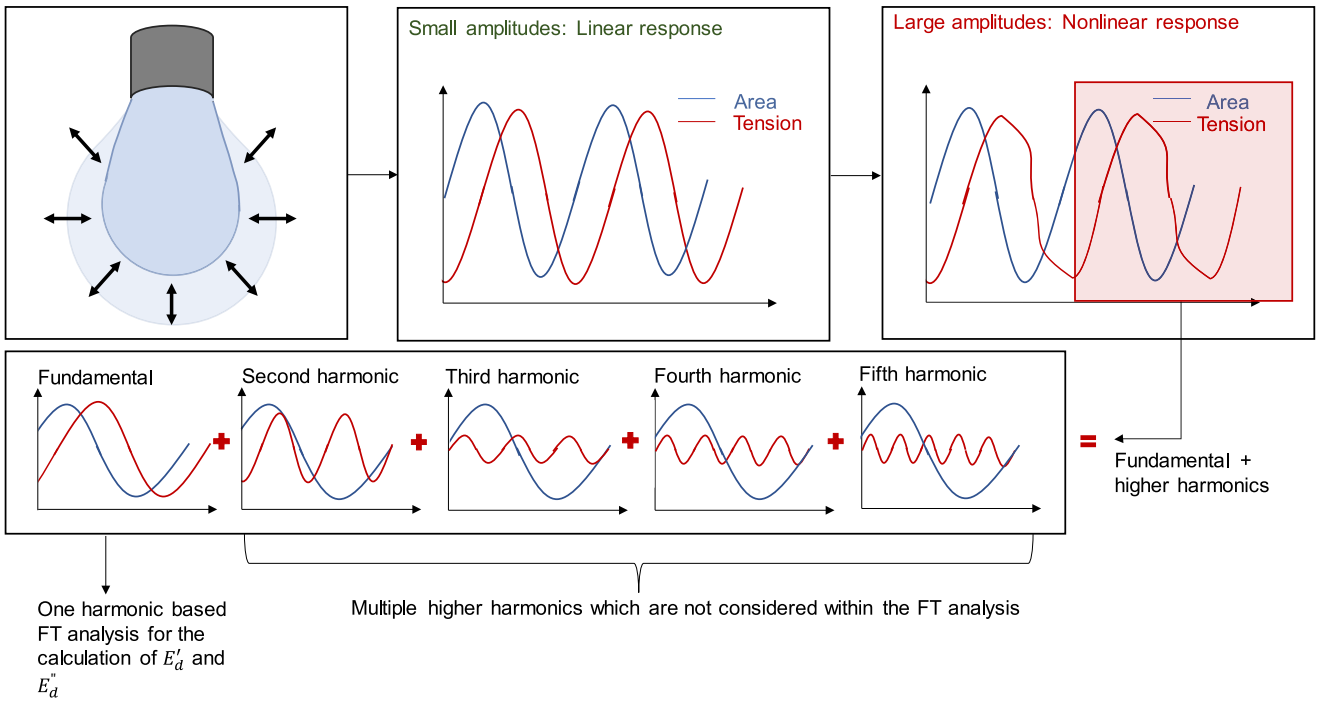


Fig. 4. Schematic overview of an oscillating droplet with the resulting area and surface tension raw signal from surface dilatational deformations at small and large amplitudes, demonstrating the contribution of higher harmonics.

analysis software can compare the contributions of the higher harmonics to the surface stress response and evaluate which higher harmonics to include to describe the surface tension signal best (e.g. by adding the third harmonic). This gives an indication of the errors in E_d' and E_d'', but the fact remains that these first harmonic-based moduli become inaccurate when the higher harmonics give a significant contribution to the surface tension signal.

A similar issue was reported for interfacial large amplitude oscillatory shear (LAOS) experiments [20,58]. Like in dilatational rheology, in LAOS a sinusoidal deformation is applied to an interface at various strains and single frequencies. For LAOS, different methods have been developed to quantify the nonlinear oscillatory stress response in the nonlinear regime, such as the use of more advanced FT-algorithms, which e.g. allow averaging the magnitude spectra so that the peaks are all positive and independent of the phase of the higher harmonics [64]. Another approach is the introduction of a nonlinear quantitative coefficient, Q, within the FT analysis to evaluate the intensity and variation of the higher harmonics compared to the first harmonic [60].

Unfortunately, these approaches cannot simply be adopted for dilatational rheology, as stress-controlled rheometers have a much lower torque limit (thus higher sensitivities) than image-based tensiometers, making the FT analysis more robust. Additionally, dilatational rheology follows slightly different mathematical and physical principles. One main difference between surface shear and dilatational rheology is the area conservation of the interface. While in surface shear rheology, the total area is conserved, this is not the case in dilatational rheology (area increases/decreases continuously during oscillation and with it, the density of the adsorbed emulsifier). Mathematically speaking, only odd harmonics are included in shear rheology because the stress response can be assumed to be of odd symmetry with respect to the directionality of shear strain or shear rate [20]. In contrast, both, even and odd harmonics appear in LAOD due to changes in density upon expansion and contraction of the interface. In other words, there is no odd symmetry, and the surface tension signal will look different in compression and expansion. Other methods are needed to quantify the nonlinear material response under LAOD deformation, which will be further discussed in Section 3.

2.5. *A more fundamental look at interfacial dynamics: The surface momentum balance*

The origin of the nonlinearities we discussed in the preceding sections may not be immediately obvious. We discuss this here in more detail by taking a closer look at the dynamics of an interface when it is deformed.

In principle, a drop tensiometer measures surface tension, and the dilatational modulus we determine by applying oscillatory area changes using Eq. (6), in which the numerator for most simple surfactant-stabilised interfaces contains only contributions from surface tension. But many surface-active components from viscoelastic solid-like (arrested) structures, after adsorption (i.e., 2D networks, soft solid glasses, or solid crystalline phases), and upon expansion/compression of these structures, deviatoric stresses are induced. From Eq. (6), it is not immediately clear how these stresses are accounted for in the measurement. These deviatoric stresses may also result in the emergence of surface shear stresses in the measured surface stress, particularly for very stiff interfaces with a high surface shear modulus. The dilatational deformation in a drop tensiometer is, in fact, not a completely all-sided deformation. In very stiff and highly elastic interfaces, the deformations in the meridional and azimuthal directions of the drop may deviate significantly, resulting in a substantial shear deformation. To see how these effects emerge in the measured surface stress signal we have to take a closer look at the Young-Laplace equation.

The Young-Laplace equation, as it is presented in Eq. (1), is in fact not a fundamental equation, but an approximation of the momentum balance at the interface [65], valid only when in-plane inertial stresses, Marangoni stresses, surface deviatoric stresses, and inertial and deviatoric stresses exerted by the adjoining bulk phases on the interface are negligible. The full momentum balance at the interface takes on the form [65]:

$$\rho^s \frac{d_s \mathbf{v}^s}{dt} - \nabla_s \cdot \mathbf{T}^s - \rho^s \mathbf{b}^s + \rho^1 \left(\mathbf{v}^1 - \mathbf{v}^s\right)\left(\mathbf{v}^1 - \mathbf{v}^s\right)\cdot \mathbf{n} - \rho^2 \left(\mathbf{v}^2 - \mathbf{v}^s\right)\left(\mathbf{v}^2 - \mathbf{v}^s\right)\cdot \mathbf{n} + \mathbf{T}^1 \cdot \mathbf{n} - \mathbf{T}^2 \cdot \mathbf{n} = 0 \tag{7}$$

Here, ρ^s is the overall surface mass density, $\mathbf{v}^s$ is the surface velocity field, ∇_s is the surface gradient operator, $\mathbf{T}^s$ is the total surface stress tensor (a tangential surface tensor), $\mathbf{b}^s$ are the body force fields per unit mass acting on the material in the interface, ρ^M ($M = 1,2$) is the bulk density evaluated at the interface for phase 1 and 2, $\mathbf{v}^M$ is the bulk velocity of the two adjoining phases evaluated at the interface, $\mathbf{T}^M$ is the stress tensor in the bulk evaluated at the interface, and $\mathbf{n}$ is the unit normal vector to the interface. The surface and bulk stress tensors are commonly split in a hydrostatic and extra stress part, according to:

$$\mathbf{T} = -P\mathbf{I} + \mathbf{S} \tag{8}$$

$$\mathbf{T}^s = \gamma\mathbf{I}^s + \mathbf{S}^s \tag{9}$$

where P is again the pressure in the bulk, $\mathbf{I}$ is the 3d unit tensor for the bulk fields, $\mathbf{S}$ is the bulk extra stress tensor, γ the surface tension, $\mathbf{I}^s$ is the surface projection tensor (the unit tensor for the tangential surface fields), and $\mathbf{S}^s$ is the surface extra stress tensor. Substituting the last two equations in the momentum balance gives:

$$\rho^s\frac{d_s\mathbf{v}^s}{dt} - \nabla_s\gamma - 2\gamma H\mathbf{n} - \nabla_s\cdot\mathbf{S}^s - \rho^s\mathbf{b}^s + \rho^1\left(\mathbf{v}^1-\mathbf{v}^s\right)\left(\mathbf{v}^1-\mathbf{v}^s\right)\cdot\mathbf{n} - \rho^2\left(\mathbf{v}^2-\mathbf{v}^s\right)\left(\mathbf{v}^2-\mathbf{v}^s\right)\cdot n + \left(\mathbf{S}^1-\mathbf{S}^2\right)\cdot\mathbf{n} + \left(P^2-P^1\right)\mathbf{n} = 0 \tag{10}$$

To arrive at this result, we have used $\nabla_s\cdot\mathbf{I}^s = 2H$, where H is the mean curvature of the interface. This balance describes the time rate of change of surface excess momentum as a function of (in order of appearance in the equation) Marangoni stresses (surface tension gradients), curvature-induced stresses, in-plane surface deviatoric stresses, body forces, inertial forces exerted on the interface by bulk phase 1 and 2, deviatoric stresses exerted by bulk phase 1 and 2, and the hydrostatic pressure difference across the interface. In general, dilatational deformations in drop tensiometry are sufficiently slow for inertial stresses to be negligible, and when viscosities of the adjoining bulk phases are low, bulk deviatoric stresses can also be ignored in the analysis of data. The equation then reduces to

$$-\nabla_s\gamma - 2\gamma H\mathbf{n} - \nabla_s\cdot\mathbf{S}^s - \rho^s\mathbf{b}^s + \left(P^2-P^1\right)\mathbf{n} = 0 \tag{11}$$

When deformations are affine, the Marangoni stress term also cancels. The second, 4th, and last term are the classical Young-Laplace equation that is used to extract the surface tension from the profile of the droplet. To see how the third term in this equation, the surface deviatoric stresses affect the surface pressure data, we first split the surface stress tensor into a contribution purely from its trace, and a symmetric traceless contribution ($\overline{\mathbf{S}^s}$):

$$\mathbf{S}^s = \frac{1}{2}(\mathrm{tr}[\mathbf{S}^s])\mathbf{I}^s + \overline{\mathbf{S}^s} \tag{12}$$

The first term on the right-hand side represents the purely dilatational stresses, the second term quantifies the surface shear stresses. Substituting this in the jump momentum balance, we obtain

$$2\gamma H\mathbf{n} + \nabla_s(\mathrm{tr}[\mathbf{S}^s]) + (\mathrm{tr}[\mathbf{S}^s])H\mathbf{n} + \nabla_s\cdot\overline{\mathbf{S}^s} + \rho^s\mathbf{b}^s - \left(P^2-P^1\right)\mathbf{n} = 0 \tag{13}$$

In standard profile analysis, only the normal component of this balance is used to extract the surface tension. The second term on the left-hand side is purely tangential, but the 4th term, representing the shear stresses, does have a normal component. Hence, the normal component reduces to

$$2\gamma_{\mathrm{eff}}H + (\nabla_s\cdot\overline{\mathbf{S}^s})\bullet\mathbf{n} + \rho^s\mathbf{b}^s\bullet\mathbf{n} - \left(P^2-P^1\right) = 0 \tag{14}$$

where

$$\gamma_{\mathrm{eff}} = \gamma + \frac{1}{2}(\mathrm{tr}[\mathbf{S}^s]) \tag{15}$$

Here, we observe that the emergence of dilatational deviatoric stresses results in an effective surface tension that comprises both a surface tension and a deviatoric stress contribution. When only image analysis is used to extract this surface tension, these two contributions cannot be separated. But when image analysis is used in conjunction with measuring the Laplace pressure, and a suitable constitutive equation is selected for the dilatational stress, these parameters can be determined individually. However, this technique termed capillary meniscus dynamometry, is not yet widely used.

The shear stress term takes on a more complicated form. Let us define a parametrisation of the interface of the form $\mathbf{z} = \mathbf{f}(y^1, y^2)$, where $\boldsymbol{z}$ denotes the position of the interface in 3D space, and $y^\alpha(\alpha = 1,2)$ are the surface coordinates. And let $\{\mathbf{a}_\alpha, \mathbf{a}_\beta\}$ be the natural basis vectors at every point on the surface, and $\{\mathbf{a}^\alpha, \mathbf{a}^\beta\}$ its associated dual basis. We can then express the shear stress with respect to these bases as $\overline{\mathbf{S}^s} = S^{\alpha\beta}\mathbf{a}_\alpha\mathbf{a}_\beta$ or $\overline{\mathbf{S}^s} = S_{\alpha\beta}\mathbf{a}^\alpha\mathbf{a}^\beta$, where $S_{\alpha\beta}$ and $S^{\alpha\beta}$ are the covariant and contravariant components of this tensor, respectively. The surface gradient of an arbitrary vector $\mathbf{c}$ is defined here as $\nabla_s\mathbf{c} = (\partial c/\partial y^\alpha)\mathbf{a}^\alpha$ and its surface divergence as $\nabla_s\bullet\mathbf{c} = (\partial\mathbf{c}/\partial y^\alpha)\bullet\mathbf{a}^\alpha$. With these definitions, we can write the normal component of the divergence of the shear stresses as ([65]

$$(\nabla_s\cdot\overline{\mathbf{S}^s})\bullet\mathbf{n} = S^{\alpha\beta}B_{\beta\alpha} = \overline{\mathbf{S}^s} : \mathbf{B} \tag{16}$$

Where $\mathbf{B}$ is the symmetric second ground form tangential tensor, equal to -$\nabla_s\mathbf{n}$ (with its trace equal to $2H$). With this, the momentum balance finally reduces to

$$2\gamma_{\mathrm{eff}}H + \overline{\mathbf{S}^s} : \mathbf{B} + \rho^s\mathbf{b}^s\bullet\mathbf{n} = \left(P^2-P^1\right) \tag{17}$$

We see that when using the standard analysis protocol of a drop tensiometer (i.e. the Young-Laplace equation), deviatoric dilatational stresses are simply lumped into the effective surface pressure extracted from the profile analysis. But, the shear stresses are not explicitly accounted for, and when shear moduli are high, this may induce significant errors in the values for the dilatational parameters. Based on the theory of 2D elasticity, E_d and the surface shear modulus G_s are related through the surface Poisson ratio ν^s [27] by

$$E_d = G_s\left(\frac{1+\nu^s}{1-\nu^s}\right) \tag{18}$$

with the Poisson ratio taking on values between 0 and 1. This means that $G_s \le E_d$ and for completely compressible interfaces ($\nu^s = 0$), G_s is at most equal to E_d. As long as $G_s \ll E_d$ contributions from shear can be neglected in Eq. (17). But recent studies have shown that when $G_s \sim E_d$, errors can be substantial when simply using Eq. (1) to extract dilatational moduli [66].

In some cases, the condition that $G_s \le E_d$ appears to be violated, and this mostly occurs in particle (i.e., Pickering) stabilised interfaces (see for example [66], and references therein). This would imply a negative Poisson ratio, and although most interfaces will have a positive Poisson ratio, technically a negative value is possible: the second law of thermodynamics requires only that both $G_s \ge 0$ and $E_d \ge 0$, which restricts the Poisson ratio to $-1 < \nu^s < 1$. A negative Poisson ratio would make the interface the two-dimensional equivalent of a meta-material. It is unlikely that a Pickering-stabilised interface would have such characteristics, and this apparent violation is more likely the result of the complex nature of these interfaces: colloidal particles are trapped by a primary fluid-fluid interface, and these particles (at sufficiently high area fraction) form a much thicker secondary interface. When the attractive interactions between the particles are weak (e.g., contact angles with the primary surface close to 90°, and/or sufficiently strong repulsive forces to screen the attraction), such that the particles do not form a strong network, a shear deformation could still result in a significant surface shear stress, because of jamming. So in shear, we would be probing the properties of the secondary interface. But particularly in

the expansion stage of dilatation, we would be expanding the primary interface, resulting in a low resistance and hence a low value for the dilatational modulus, which is much smaller than the surface shear modulus. In compression, we would again expect jamming, and hence a considerably higher surface stress than in expansion. But this type of interface often displays buckling even at moderate strains, which makes further analysis of the dilatational response impossible.

To summarise, for complex interfaces, shear parameters should always be tested along with the dilatational properties to determine whether shear effects are small enough to be neglected. If this is not the case, dilatational deformations should be limited to small amplitudes, or one should resort to different methods, such as the quadrotrough [33], which can apply a purely all-sided dilatational deformation even for highly elastic solid-like interfaces.

The deviatoric stresses in the effective surface tension are not the only source of nonlinearities in the response. The surface tension itself is dependent on the surface load of the adsorbed component through an equation of state, which is in general a nonlinear equation, even for the simplest types of interfaces, i.e., interfaces described by the 2D ideal gas law. The surface tension will increase upon expansion and decrease upon compression. As a result of the nonlinearity of the equation of state, when deformations are sufficiently large, these effects are not symmetric with respect to the nondeformed state, leading to the emergence of the even harmonics, which we mentioned in the previous section.

3. Qualitative and quantitative analysis of LAOD

3.1. Lissajous plots

In view of our discussion in the previous section, other methods are needed for the analysis of the nonlinear material response in LAOD deformations. A popular qualitative analysis of LAOD is by plotting the measured cyclic surface stress/pressure directly against the time-dependent area signal in Lissajous(-Bowditch) plots ([20,50,67,46]). In these plots, the surface stress changes in a clockwise direction, and at sufficiently small amplitudes, the plots will be point-symmetric with respect to the origin (Fig. 5). Here, the upper part of the plot is the expansion part of the deformation cycle, where the area changes from its minimum to its maximum value. The opposite is true for the bottom part of the plot, which represents the compression part of the deformation cycle. The shape of the Lissajous plots showcases the rheological response. In a purely elastic response, the stress is in phase with the amplitude, resulting in a straight line in the plots. In a purely viscous response, the stress response is 90° out of phase with the applied amplitude, creating a circle in the plots. Viscoelastic behaviour is shown as a combination of both elastic and viscous signals, in the form of an ellipsoidal shape. In the NLVE regime, Lissajous plots will show a distortion from the original symmetric shape where the plots can show different responses in expansion or compression as a result of cyclic interfacial strain stiffening and softening, as shown in the example, where Lissajous plots become more asymmetric at higher deformations (Fig. 5).

The strain stiffening and softening visually shown in Lissajous plots was, for instance, quantified in Van Kempen et al. [46], where the intracycle strain stiffening ratio in compression S_{com} and expansion S_{exp} was analysed, in the same way as [20] defined the intracycle strain stiffening ratio S for bulk rheology. Such a method is a geometrical interpretation calculated from the shape of Lissajous curves, and the higher harmonic components obtained via Fourier transformation of the oscillatory signal are not used. A more accurate method to quantify the nonlinear contributions in LAOD, a general stress decomposition (GSD) method was developed by de Groot et al., [25], and is described in the next section.

3.2. General stress decomposition

The general stress decomposition (GSD) assumes that the total surface stress response in oscillatory dilatation results from two contributions: 1) network stresses due to network interactions + linear surface density changes and 2) nonlinear stress contributions due to surface density changes. The network contributions are similar to those found for isotropic shear rheological responses, but the contributions due to surface density changes are unique to interfacial dilatational rheology. The combination of network and nonlinear surface density changes yields an asymmetric response in compression and expansion in large amplitude oscillatory dilatational rheology (Fig. 6A). This surface stress response, $\Pi(t) = \gamma(t) - \gamma_{eq}$, where again γ_{eq} is the surface tension of the equilibrium or nondeformed interface, can be described by:

$$\Pi(t) = \sum_{k=0}^{\infty} q_k sin(k\omega t + \delta_k) \tag{19}$$

Here, q_k is the amplitude of the kth harmonic, ω is the base frequency, t is time, and δ_k is the phase shift of the kth harmonic. The surface stress $\Pi(t)$ should not be confused with the surface pressure Π_0, which is defined with respect to the surface tension of the bare interface. The latter is always positive, whereas $\Pi(t)$ can take on positive and negative values. Note that here, also the even multiples of the applied frequency are included as opposed to only the odd ones in shear experiments [20]. The overall stress ($\Pi(t)$), is then split into τ_1 and τ_2 that contain all the odd harmonic contributions, and τ_3 and τ_4 which are composed of all the even harmonics. Their form is given by:

$$\tau_1 = \sum_{k=0}^{\infty} b'_{2k+1} sin((2k+1)\omega t) \tag{20}$$

$$\tau_2 = \sum_{k=0}^{\infty} a'_{2k+1} cos((2k+1)\omega t) \tag{21}$$

$$\tau_3 = \sum_{k=0}^{\infty} c'_{2k} sin(2k\omega t) \tag{22}$$

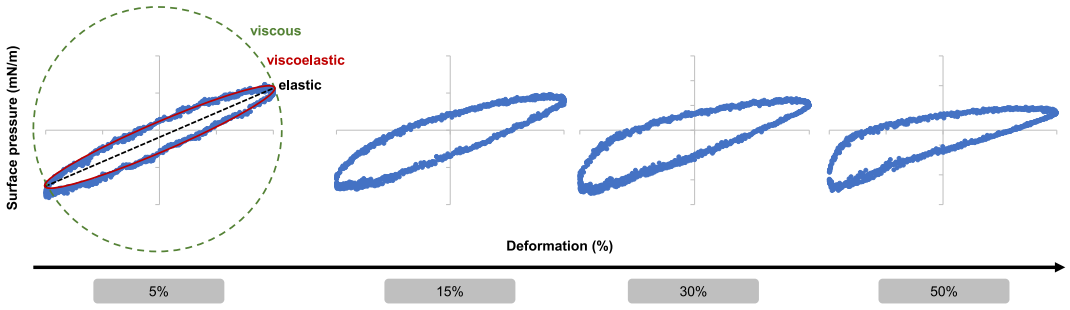


Fig. 5. Example of Lissajous plots from small to large deformations (5–50 %) of a whey protein-stabilised air-water interface. The left panel shows examples of a purely elastic, viscoelastic and a purely viscous plot.

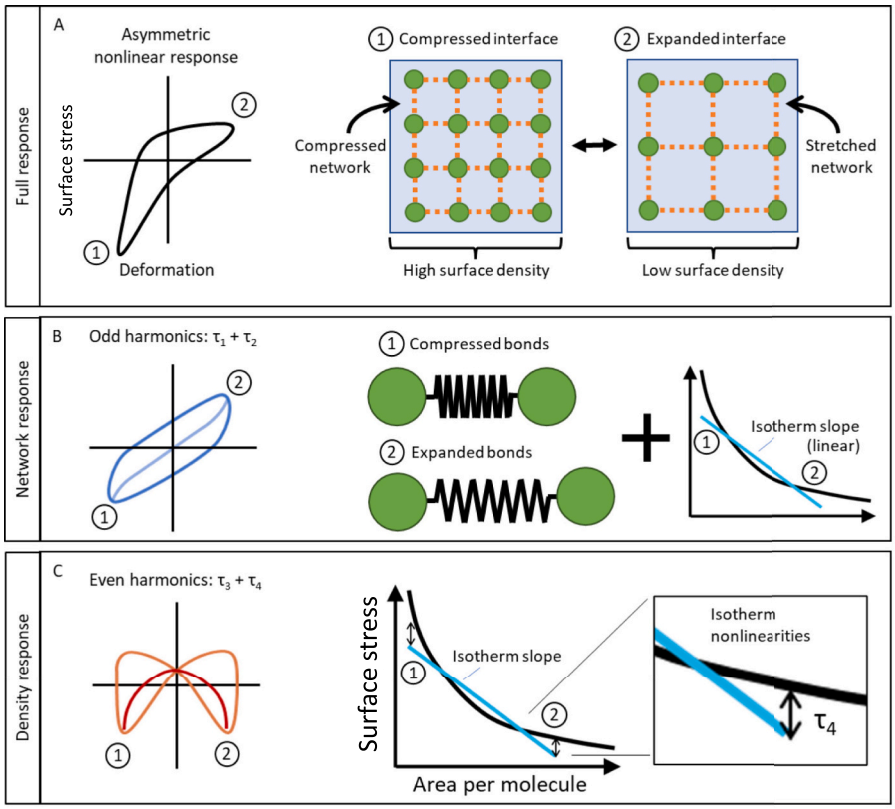


Fig. 6. Decomposition of an asymmetric nonlinear response (A) into the odd harmonics (B) and the even harmonics (C). The right side shows a schematic representation of the underlying mechanism for establishing the stress response in compression (1) and expansion (2). The compressed interface contains more proteins per surface area and a compressed network; The expanded interface contains less proteins per surface area and a stretched network.

$$\tau_4 = \sum_{k=0}^{\infty} d'_{2k} cos(2k\omega t) \tag{23}$$

The coefficients a'_{2k+1}, b'_{2k+1}, c'_{2k}, and d'_{2k} are the amplitudes for each harmonic contained in respectively τ_1 - τ_4. For symmetric viscoelastic responses, such as those obtained in surface shear rheology or small amplitude dilatational rheology, only τ_1 and τ_2 are present. Here τ_1, describes the elastic contribution and τ_2 the viscous contribution is similar to the framework provided by [20]. Asymmetric responses (Fig. 6A), such as those obtained in LAOD, contain all four components. Here, τ_3 and τ_4 represent the energy-dissipating and energy-storing contribution. In Fig. 6A, an asymmetric response is decomposed into odd and even harmonics. The odd harmonics are point symmetric around the origin, e.g. the stress response to compression and expansion is equal but opposite. The example of compressing a spring gives some intuition for the matter. The spring can display nonlinearities but must have an opposite response in compression and expansion (Fig. 6B). Nonlinearities in the odd harmonics display strain stiffening or softening behaviour and strain rate thinning and thickening behaviour of the interface.

These network nonlinearities are typically related to deviatoric stresses generated by changes in the interfacial microstructure that change the effective surface tension (γ_{eff}). It is important to note that the first harmonic shows the linear portion of the response and therefore contains both a linear network (or deviatoric) contribution and the linearised portion of the response due to surface density changes. The even harmonics of the intracycle surface density changes result in an identical response in compression and expansion, i.e., the plots are symmetric with respect to the vertical axis (Fig. 6C). As mentioned above, the nonlinearities in the even harmonics stem from the nonlinearity of the surface pressure area isotherm. For the simplest case, the ideal surface gas, this curve is smoothly decreasing with increasing surface area ($\partial\Pi_0/\partial A < 0$ & $\partial^2\Pi_0/\partial A^2 > 0$) as shown in Fig. 6C. This plot shows that large compression and expansion lead to an identical deviation from the linear response, which ends up in the even harmonics (τ_3 and τ_4). Here, τ_4 shows that the recoverable (elastic) contribution of nonlinearities in the surface pressure isotherm and τ_3 the dissipative (viscous) contribution.

The availability of this decomposition also makes it possible to analyse the intracycle nonlinear viscous properties by plotting strain rate versus τ_2, creating a so-called viscous Lissajous plot. This can provide additional information on interfacial strain thickening or thinning.

We should however emphasise that the validity of the assumption that odd harmonics represent network interactions and even harmonics are related to density changes relies on a number of assumptions: firstly, the interfacial microstructure should be isotropic, as anisotropic network structures may generate even harmonics [68]. Additionally, the surface pressure area isotherm should be a smoothly decreasing function with increasing area; otherwise, the surface density changes may generate odd harmonics. This may occur when the interface undergoes a phase transition during compression or expansion, which can lead to inflexion points in the surface pressure isotherm. The isotropy of interfacial structures can be verified with microscopic techniques such as Langmuir-Blodgett deposition in combination with atomic force microscopy. Verification of the surface pressure-area isotherm is more challenging as deformation in LAOD is often performed around a dynamic equilibrium as opposed to the near-equilibrium conditions when measuring surface pressure area isotherm in a Langmuir trough. Many surfactants are known to go through phase transitions under compression [69,70,71]. However, it is expected that this only occurs at extremely large deformations for specific surfactants [72,70]. Nevertheless, this assumption seems acceptable for many surface-active agents, particularly at air-liquid interfaces. At oil-water interfaces, the

decomposition may break down at large amplitudes, often as a result of buckling of the interface. We will explain this in more detail when we discuss results for specific systems (section 4).

3.3. Quantification of Lissajous plot data

The general stress decomposition now provides an opportunity to define new parameters for interfacial dilatational stress responses to facilitate the analysis of a complete amplitude sweep. In Fig. 7, we provide a visual aid showing each quantification parameter in its corresponding Lissajous plot.

For the stress, τ_1, parameters similar to those used in shear rheology can be introduced, where $E_{\tau 1L}$ is defined as the secant modulus, i.e., the slope of the line connecting the origin and the point in the curve at maximum expansion; $E_{\tau 1L}$ describes the overall stiffness of an interface, and $E_{\tau 1M}$ the stiffness at zero strain, they are given by:

$$E_{\tau 1L} = \frac{\tau_1}{\varepsilon}\Big|_{\varepsilon=\pm\varepsilon_0} = \frac{\sum_{k=0}^{\infty} b'_{2k+1}(-1)^k}{\varepsilon_0} \tag{24}$$

$$E_{\tau 1M} = \frac{d\tau_1}{d\varepsilon}\Big|_{\varepsilon=0} = \frac{\sum_{k=0}^{\infty} (2k+1) b'_{2k+1}}{\varepsilon_0} \tag{25}$$

Where, ε is the strain and ε_0 is the maximum strain (related by $\varepsilon = \varepsilon_0 \sin(\omega t)$). Comparing the decomposition of dilatational stress to shear stress reveals that $E_{\tau 1L}$ and $E_{\tau 1M}$ are equivalent to G_L' and G_M' as defined by [20].

Both describe the stiffness of the interfacial network, and any differences between the two give an indication of the effects of deformation on the stiffness of the interface. This can be expressed by the S-factor, a measure for intracycle strain stiffening or softening, which is given by:

$$S = \frac{E_{\tau 1L} - E_{\tau 1M}}{E_{\tau 1L}} \tag{26}$$

Next, τ_4 is also quantified with a modulus, $E_{\tau 4}$ (Eq. 9), which describes the contributions to the overall stress due to surface density changes: a lower (more negative) $E_{\tau 4}$ relates to a more significant contribution of the surface density changes to the intracycle surface stress. Additionally, the overall shift of the surface stress is quantified by γ_s (Eq. 10) which displays how far from equilibrium the deformations are performed. $E_{\tau 4}$ and γ_s are given by:

$$E_{\tau 4} = \frac{\tau_4(\varepsilon_0) - \tau_4(0)}{\varepsilon_0} = -\frac{\sum_{k=0}^{n} 2d'_{4k+2}}{\varepsilon_0} \tag{27}$$

$$\gamma_s = \gamma_{eq} - \gamma_0 = d'_0 \tag{28}$$

Since, τ_2 and τ_3 are dissipative contributions, a dissipated energy per cycle can be calculated for both. These dissipated energies correspond to the enclosed areas of their respective loops, denoted by $U_{d\tau 2}$ and $U_{d\tau 3}$, and are calculated using:

$$U_{d\tau 2} = \oint \tau_2(t)\dot{\varepsilon}(t)\, dt = \pi\, \varepsilon_0^2\, E''_1 \tag{29}$$

$$U_{d\tau 3} = \oint \tau_3(t)\, \dot{\varepsilon}(t)\, dt = 2\varepsilon_0^2 \sum_{k=1}^{n} \left(\frac{E_{2k\tau 3} * k}{k^2 - 1/4} \right) \tag{30}$$

Here, $\dot{\varepsilon}(t)$ is the dilatational strain rate, E''_1 is the loss modulus from the first harmonic and $E_{2k\tau 3} = c'_{2k}/\varepsilon_0$ is the loss modulus from the even harmonics of the Fourier series. Their sum is equal to the total dissipated energy per cycle, and equal to the area enclosed by the overall loop. To obtain the coefficients a'_{2k+1}, b'_{2k+1}, c'_{2k}, and d'_{2k} in eqs. 20–23, a Fast Fourier Transform (FFT) is applied to the time-dependent surface stress signal. This approach will decompose the signal into its harmonic components, of which the amplitude and phases of the relevant Fourier coefficients are then extracted. These coefficients are then directly used in the expressions to obtain τ_1–τ_4, allowing the reconstruction of the individual stress contributions over the whole deformation cycle. This same approach is used to generate the Lissajous plots in Fig. 7, which visualises the decomposed stress signals for each of the contributions, and, finally, allowing quantification. For a step-by-step description of the method and theoretical background, we refer to the original GSD article [25].

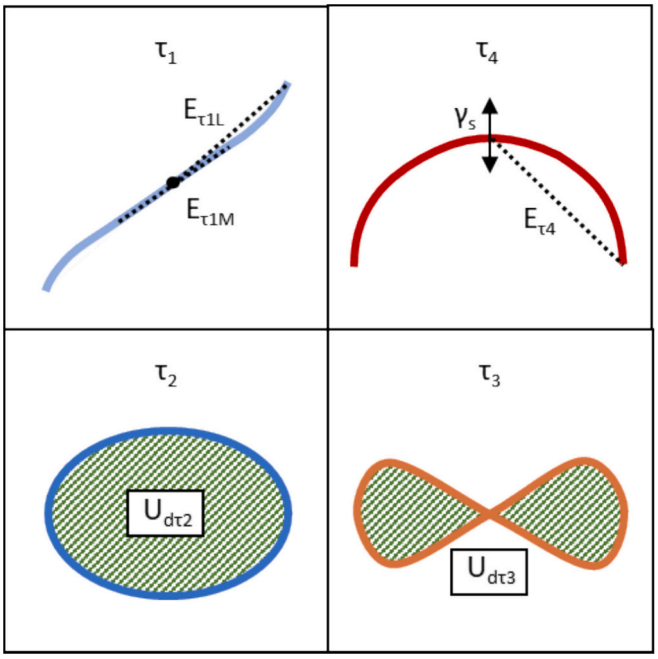


Fig. 7. Quantification of each decomposed stress τ_1 – τ_4, adapted from [25]. The horizontal axis represent deformation and the vertical axis represents surface stress.

3.4. Interpretation of Lissajous plots

Based on the separation into odd and even harmonics, we explore the phase space of interfacial viscoelastic responses in Fig. 8. Here, Lissajous plots are shown as a function of the intensity of the elastic component of the second and third harmonic, I'_2 and I'_3; their ratio and intensity can vary depending on the surface-active component and applied strain. Surface-active components without any attractive in-plane interactions will behave like colloidal glasses under deformation, and only surface density changes will affect the intracycle stress. These colloidal glass-like interfaces will, therefore, solely contain nonlinearities related to the second (or higher even) harmonic(s) and show Lissajous plots similar to the ones on the horizontal axis, where $I'_3 = 0$. Contrarily, the Lissajous plots on the vertical axis represent a gel-like structure with strong attractive in-plane interactions and limited surface density changes or a linear surface pressure area-isotherm. Depending on whether the gel network is strain-softening or strain-stiffening, the respective I'_3 will be positive or negative. Quantification of these different contributions can be done with Fourier-transform rheology, similar to what is done in bulk rheology [73]. Here, the intensity of each harmonic is normalised for the first harmonic and fitted with the empirical scaling law:

$$I_{n/1} = \frac{I_n}{I_1} = p_n \varepsilon_0^{-1} + q_n \varepsilon_0^{n-1} \tag{31}$$

were $I_{n/1}$ is the normalised intensity of the corresponding harmonic n, and p_n and q_n are scalar coefficients. At large deformations, q_n dominates this equation and is essentially an intrinsic nonlinearity [73]:

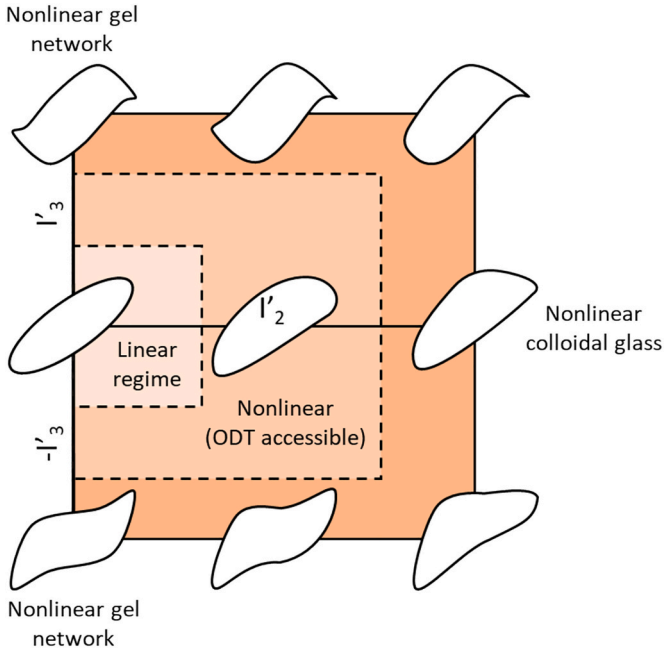


Fig. 8. Typical Lissajous plots for varying contributions of the elastic component of the second harmonic (I'_2) versus the elastic component of the third harmonic (I'_3). The extremes of only third harmonic contributions (which lie on the vertical axis) show nonlinear behaviour similar to that observed for a gel network in shear rheology, and the extreme of only second harmonic contributions shows wide and asymmetric plots we typically observe in dilatational rheology. Negative I'_2 are not shown since they are rarely observed in literature and so far these lack physical interpretation. The shaded colours roughly indicate typical boundaries of the linear regime, the nonlinear regime accessible by an oscillating drop tensiometer, and inaccessible part of the nonlinear regime; boundaries of these regimes may vary depending on the device and the surface-active component.

$$q_n = {}^nQ_0 = \lim_{\varepsilon_0 \to 0} \frac{I_{n/1}}{\varepsilon_0^{n-1}} \tag{32}$$

The relative values of the intrinsic nonlinearity of the odd versus even harmonics give an indication of the nature of interface stabilisation, and indicate whether the nonlinear response of an interface is dominated by network disruptions or density changes. This has been used to show that casein micelles form stiffer interfaces at highly polar oils through particle-like interface stabilisation [74].

For the examples of Lissajous plots with varying contributions of network and density nonlinearities in Fig. 8, the previously described parameters help to quantify the changes in viscoelasticity over the course of an amplitude sweep. The overall interfacial stiffness is described by $E_{\tau 1L}$ and the initial stiffness by $E_{\tau 1M}$. In the presence of an interfacial network which is progressively disrupted with increasing strain amplitude, the values of $E_{\tau 1L}$ will decrease with increasing amplitude. In the absence of a network, the interfacial stiffness will remain constant as there is no network that can break down under large deformation amplitude. Additionally, the total energy dissipation within one cycle can be calculated with $U_{d\tau 2}$ to quantify the viscous energy dissipation and describe the flow behaviour of an interface, e.g. microstructural rearrangements. The even harmonics are quantified with similar parameters, and the stiffness regarding nonlinearities in the surface density changes is quantified by $E_{\tau 4}$. $E_{\tau 4}$ has a more negative value for surface-active components with large changes in surface density, i.e., components that exhibit limited compensation of the density change by exchange with the bulk phase and a highly nonlinear and steep surface pressure-area isotherm. The viscous dissipation related to surface density changes is quantified by $U_{d\tau 3}$. The shift, γ_s, indicates a baseline shift, and is a measure of how far the oscillations are driving the interface out of equilibrium. Generally, this shift is negative. Physical interpretation of the dissipated energy and shift described by $U_{d\tau 3}$ or γ_s remains challenging, and additional research should be conducted.

4. Application of LAOD to investigate the nonlinear deformation behaviour of various emulsifiers – Examples

In this section, we will show the potential of LAOD in elucidating the mechanical properties of liquid-liquid or air-liquid interfaces. LAOD can give insights into the type of interfacial microstructures formed and the type of in-plane interactions between adsorbed interfacial stabilisers. We will discuss several categories of stabilisers, such as low molecular weight surfactants, proteins and particles.

LAOD is especially powerful in combination with microstructure imaging, as the latter can underpin the microscopic structural features in interfacial films suggested by the LAOD results. Examples are Langmuir-Blodgett or – Schaefer deposition of interfacial layers onto substrates, followed by atomic force microscopy (AFM) or electron microscopy ([75,76]; Q. [77,14]). Less invasive and more in-situ methods include Brewster angle microscopy (BAM) in combination with a Langmuir through or multiphoton microscopy [78,79,80]. Examples where LAOD and imaging have been used for different emulsifiers such as surfactants, proteins and particles are shown in Table 1. A category of mixtures is also shown, as LAOD has been proven useful in underpinning the role of different emulsifiers in mixed systems.

4.1. Surfactants

Low molecular weight surfactants are generally amphiphilic molecules with a well-defined hydrophilic headgroup and hydrophobic tail. In literature, most surfactants are not expected to self-assemble into interfacial 2D microstructures and are, therefore, assumed to form weak and highly viscous interfacial films which are unstable under large deformations. At low deformation rates, their dilatational properties are largely governed by dynamic exchange of the surfactant between interface and the bulk solution. However, under fast deformations, the exchange of surfactant between interface and bulk cannot keep up with the deformation, as described by the Lucassen-van den Tempel theory [53]. In such cases, the interfacial response appears elastic not because of structural rigidity, but due to the inability of the adsorbed surfactant molecules to rapidly move to the bulk phase. Nevertheless, recent studies have shown that some surfactants (e.g. phospholipids, saponins) may form viscoelastic networks at the interface with complex nonlinear behaviour [102,103,104,105,106].

So far, only a few studies have used GSD to analyse the interfacial behaviour of surfactant-stabilised interfacial films. For example, De Groot et al. reported nonlinear in-plane behaviour for lecithin. Here, the GSD showed that in-plane network interactions are nearly absent when deforming lecithin-stabilised air-water interfaces, and the surface density changes dominated the overall stress signal [25]. Risse et al. expanded the investigation of the nonlinear behaviour of lecithin, which in fact are phospholipid mixtures, and showed that unsaturated phospholipids lack in-plane interactions at the oil-water interface. In contrast, saturated phospholipids formed stiff interfacial films with dominating in-plane network interactions [49]. The difference between unsaturated and saturated phospholipids comes from the wider fatty acid tails of the unsaturated phospholipids that provide steric hindrance at the interface, resulting in less tight packing of the head groups at the interface. Similar behaviour was also shown for the surfactants Tween, Span and Brij, which have different head- and tail-group sizes [83].

Escin is a triterpenoid bidesmosidic (2 sugar residues) saponin which forms stiff air-water interfaces with dilatational (and shear) viscoelastic moduli of 1 to 2 orders of magnitude larger than those of proteins at the air-water [72,107,104,95]. The stiff interfacial films resulted in non-Laplacian droplet shapes at larger deformations, which were reflected in highly nonlinear Lissajous plots. The Lissajous plots of escin showed a

Table 1
Literature overview on works that have performed LAOD. Abbreviations: air-water (A-W), oil-water (O-W), elastic storage modulus (E_d'), elastic loss modulus (E_d''), general stress decomposition (GSD).

Emulsifier		Interface type	Nonlinearity analysis method	Main findings using LAOD	Reference
Surfactants	Escin	A-W	Lissajous plots	Escin formed incredibly stiff glassy-like interfaces, leading to the loss of the Laplacian shape of the droplet at higher deformations. Highly nonlinear response was not analyzable using moduli. Lissajous plots showed buckling and multi-layer formation of the interface upon compression.	[72]
	Lecithin	A-W	E_d' and E_d'' moduli, Lissajous plots, GSD	Lecithin shows complex nonlinear behaviour in Lissajous plots. GSD showed that in-plane interactions are very weak for lecithin, as the surface density changes dominate the overall stress response.	[25]
	Oligofructose	A-W	Lissajous plots	Oligofructose fatty esters show highly nonlinear behaviour, only visible when performing LAOD and depicting results in Lissajous plots.	[46]
	Span 65	A-W	E_d' and E_d'' moduli	Molecules self-assemble into a viscoelastic network.	[81]
	Asphaltene	O-W	E_d' and E_d'' moduli, Lissajous plots	Coal asphaltenes formed very stiff interfaces, resulting in buckling and multi-layer formation at the interface upon compression, similar to Escin. Petroleum and bitumen asphaltenes, which contain more aliphatic groups, formed more deformable interfacial layers.	[82]
Proteins	Tween 20/60/80, Span 20/60/80, BrijS20	O-W	E_d' and E_d'' moduli, Lissajous plots	Emulsifiers were crystallised by the heat-cool cycle. First-harmonic-based moduli were unreliable; Lissajous plots showed how the interfacial stiffness increased with higher FA chain lengths.	[83]
	(Un)saturated phospholipids	O-W	E_d' and E_d'' moduli, Lissajous plots, GSD	Unsaturated phospholipids formed predominantly viscous interfacial layers (E_d'' > E_d'). Saturated phospholipids formed stiff interfaces, and GSD showed that network interactions dominate.	[49]
	Rapeseed proteins – napin and cruciferin	A-W	E_d' and E_d'' moduli, Lissajous plots	Napin formed thinner and weaker interfacial layers than cruciferin. Lissajous plots showed that cruciferin dominated napin-cruciferin mixtures.	[84]
	Rubisco proteins	A-W	E_d' and E_d'' moduli, Lissajous plots, GSD	Rubisco proteins were extracted using two methods. The moduli showed only minor differences, but the GSD revealed that less aggregated proteins lead to more network-dominated interfaces.	[85]
	Wheat aqueous phase proteins	A-W	E_d' and E_d'' moduli, Lissajous plots, GSD	Wheat aqueous phase proteins were fractionated using ethanol. LAOD and GSD analysis showed that the ethanol-soluble fraction dominates the interfacial mechanics of the whole fraction.	[86]
	Ovalbumin	A-W	E_d' and E_d'' moduli, Lissajous plots	Ovalbumin proteins were tested at different pH and salt concentrations. Conditions with lower protein charge led to increased interactions between adsorbed proteins, shown using LAOD.	[87]
	Casein proteins	O-W	E_d' and E_d'' moduli, Lissajous plots, GSD	The stiffness of interfacial layers formed by casein proteins was determined by their ability to bind calcium ions, which could only be shown by differentiating between network interaction or surface density contributions using the GSD.	[88]
	Lipase	O-W	E_d' and E_d'' moduli, Lissajous plots	Lissajous plots revealed that enzymatic hydrolysis using lipase on oil leads to stiff, highly nonlinear interfaces, which were not visible when only first-harmonic-based moduli were used.	[89]
	Potato peptides	O-W	E_d' and E_d'' moduli, Lissajous plots	Rheology showed how a higher exposed β-sheet content leads to interfacial stiffness, as shown by LAOD.	[90]
	Whey protein	O-W	E_d' and E_d'' moduli, Lissajous plots	Whey proteins formed stiff viscoelastic solid-like networks.	[18]
	Pea proteins	O-W	E_d' and E_d'' moduli, Lissajous plots, S-factor	LAOD rheology revealed that the overall pea protein interfacial behaviour is dominated by the protein types known as 7S and 11S.	[91]
	Pea proteins	O-W	Lissajous plots	Pea proteins were extracted with endogenous phospholipids. The work showed using LAOD how these phospholipids play a major role in pea protein oil-interface formation, thereby largely reducing interactions between adsorbed pea proteins.	[92]
	Casein and tannic acid	A-W	E_d' and E_d'' moduli, Lissajous plots, S-factor	The formation of casein and tannic acid complexes increased the interfacial elasticity compared to casein alone, and a soft glassy interface was formed.	[93]
	Rapeseed proteins and oleosomes	A-W	E_d' and E_d'' moduli, Lissajous plots	Rapeseed proteins formed stiff interfacial films, while oleosomes formed weak stretchable layers. In mixture, oleosomes can weaken the rapeseed protein film. Lissajous plots showed how, by controlling the rapeseed protein concentration, the impact of oleosomes can be reduced.	[94]
Mixtures	Whey protein and escin	A-W	E_d' and E_d'' moduli, Lissajous plots	Escin formed much stiffer interfaces than whey proteins. Replacing only 12.5 % of protein by escin leads to substantially stiffer interfaces.	(J. [95])
	Whey and pea protein	A-W O-W	E_d' and E_d'' moduli, Lissajous plots	Whey proteins formed stiffer interfacial layers than pea proteins. LAOD showed how a mixture of both proteins leads to an intermediate behaviour.	[75]
	Membrane proteins and phospholipids	O-W	E_d' and E_d'' moduli, Lissajous plots, GSD	GSD revealed that phospholipids contributed to the network stiffness by increasing the surface density contribution, while the membrane proteins were responsible for network interactions.	(J. [96])
	Oleosome membrane and curcumin	O-W	E_d' and E_d'' moduli, Lissajous plots, GSD	GSD revealed how curcumin enhances molecular interactions between oleosome membrane molecules, forming a stiff glassy-like structure.	[97]
Particles	Cellulose nanocrystal	A-W	E_d' and E_d'' moduli, Lissajous plots	Salt addition screens charges, leading to interactions between nanocrystals. The result is an increased interfacial stiffness.	[98]
	Gliadin nanoparticles	A-W	E_d' and E_d'' moduli, Lissajous plots, GSD	LAOD was started at different ageing times, showing a transformation of the nanoparticles at the interface. The particles slowly deformed and merged into a viscoelastic 2D solid film.	[23]
	Soy protein fibrils	A-W	Lissajous plots, S-factor	LAOD revealed that fibrils did not contribute to interface stabilisation, as the free peptides dominated the air-water interface.	[99]
	Whey protein nanoparticles	A-W	E_d' and E_d'' moduli, Lissajous plots, S-factor	These nanoparticles were not surface active, and the nonaggregated material was found to dominate the air-water interface stabilisation.	[19]

(*continued on next page*)

Table 1 (*continued*)

Emulsifier	Interface type	Nonlinearity analysis method	Main findings using LAOD	Reference
Soy protein coacervate particles	O-W	Lissajous plot	Soy protein coacervates formed extremely stiff interfacial films that showed wrinkling upon compression at lower pH values. This behaviour was successfully captured using Lissajous plots.	[100]
Chitosan Microgel	O-W	E_d' and E_d" moduli, Lissajous plots, S-factor	LAOD rheology revealed how chitosan microgel particles form stiffer interfacial films upon swelling, which can be induced by varying the pH and particle production method.	[101]

crossover (Fig. 9), which is the result of a combination of interfacial buckling, fracturing, and forming multilayers upon compression, which then return to a stretched state upon expansion (J. [95]). Similar high nonlinearity was also observed for quillaia saponin at the oil-water interface, using LAOD [108]. Escin has these capabilities due to its molecular structure, as it possesses hydrophobic aglycone parts that interact via attractive van der Waals forces, and sugar residues that are responsible for dipole-dipole attraction and hydrogen bonds [107,106]. Asphaltene is another category, while not being classified as surfactants, it may exhibit interfacial activity under specific conditions. As asphaltene can form similar types of stiff interfacial films as the saponins due to the interaction between aliphatic and aromatic parts of the asphaltenes at the interface [82].

4.2. Proteins

Proteins are biopolymers, of which, many are surface-active. Surface active proteins can adsorb at the interface, followed by structural rearrangements and interactions, finally, forming viscoelastic films [109,10,110]. The type of interface formed depends largely on the type of interactions and interfacial arrangements of adsorbed proteins, which can be effectively analysed using LAOD [89,85,18,84]. An example was shown for rapeseed proteins, where air-water interfacial films of rapeseed proteins, specifically napin and cruciferin, were studied [84]. Napin is a smaller more hydrophilic protein, which forms thinner and weaker interfacial structures. Cruciferin proteins, which are about 20 times larger than napin and more hydrophobic, can form stiffer interfacial layers. This also resulted in a higher stability of cruciferin-stabilised foams than napin-stabilised ones. These insights were obtained using a combination of LAOD rheology and microstructural imaging of the interface. Examples of atomic force microscopy images of Langmuir-Blodgett films of cruciferin and napin are shown in Fig. 10. Here, one can observe a heterogeneous structure for cruciferin, suggesting a gel-like structure, while napin showed a thinner and more homogenous structure. The combination of interfacial rheology and imaging has been useful in explaining interfacial microstructural differences between proteins, such as dairy proteins, plant proteins, and peptides [18,111,112,113,80].

The GSD has been applied for several protein-stabilised interfaces (e.g., rubisco and casein proteins). Ma et al. extracted rubisco proteins from spinach using two different methodologies [85]. Proteins that were mildly extracted retained their native structure, and these proteins were

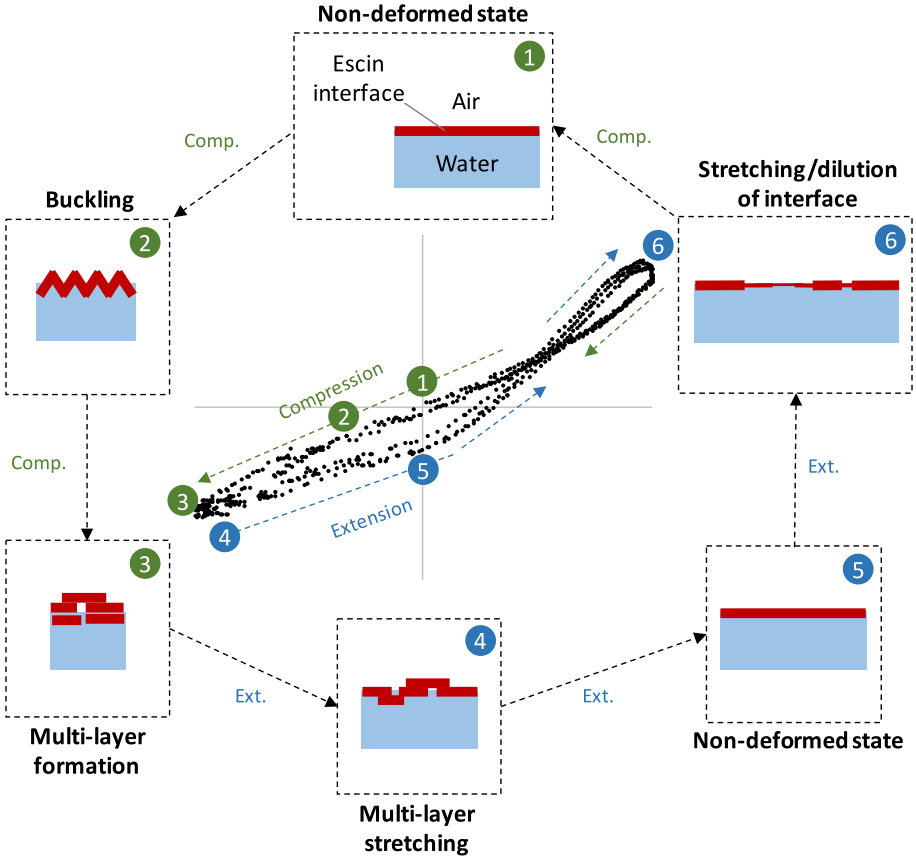


Fig. 9. A Lissajous plot of an interface stabilised by escin at 5 % deformation with schematic drawings of potential phenomena upon deformations based on work of Giménez-Ribes et al. *[72]*.

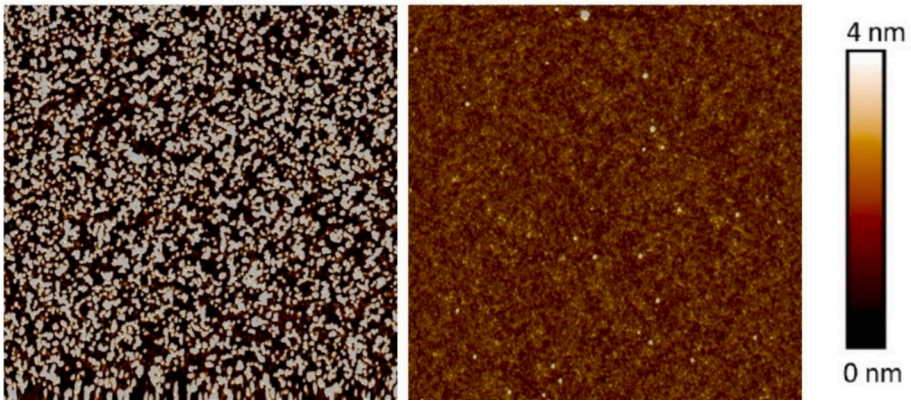


Fig. 10. Atomic force microscopy (AFM) images of Langmuir-Blodgett films created from cruciferin (left) and napin (right) air-water interfaces at a compression state of 23 mN/m [84].

able to form stiffer interfacial films. Using the second extraction method, the Rubisco proteins were heavily aggregated and unable to interact at the interface, resulting in weak interfaces. This highlights the importance of protein molecular state on interfacial interactions. De Groot et al. showed how different casein proteins interact with calcium ions at the interface. Proteins with a strong affinity for calcium were able to form stiffer interfaces [88]. In both cases, the GSD gave quantitative insights, showing when network interactions or surface density changes dominated the interfacial properties.

Another key property that drives protein interfacial interactions seems to be the protein secondary structure. A recent study combined LAOD rheology and synchrotron radiation circular dichroism to characterise oil-water interfaces and emulsions stabilised by potato peptides with varying α-helix, β-sheet, and random coil compositions [90]. Peptides that are high in β-sheet content formed stiff solid-like interfacial films, while peptides without these sheets formed weak stretchable interfaces.

4.3. Mixtures

LAOD has been proven useful in understanding the role of individual components in multicomponent systems, such as protein-protein and protein-surfactant mixtures. When both stabilisers are initially present in the bulk phase, this may lead to competitive adsorption (one stabiliser dominates the interface), co-adsorption (both adsorb forming homogenous or heterogenous blends at the interface), and/or displacement (one stabiliser replaces the other at the interface over time) [114]. For instance, caseinate-tannic acid mixtures formed stiffer interfaces than caseinate and tannic acid alone [93]. The molecules interacted and formed complexes that co-adsorbed, where the tannic acid probably further connected the proteins after adsorption, yielding stiffer interfaces. Co-adsorption without interactions is also possible, as shown for whey protein and pea protein mixtures, where both proteins adsorbed onto the interface, and hindered each other's network formation [75]. Sequential adsorption can be studied by first forming an interface in a drop tensiometer with the first stabiliser, followed by the exchange of the water or oil phase, and introducing a second stabiliser. Performing LAOD before and after the exchange gives insights into the potential displacement of the first stabiliser by the second one [115].

Interfaces stabilised by mixtures of proteins and smaller surfactants are especially suitable for analysis using LAOD. The first example is a whey protein-escin mixture, where, at low escin concentrations, the escin can increase the stiffness of the mixture ([107]; J. [95]). At high escin concentrations, escin will dominate the interfacial behaviour of the mixtures. The opposite is also possible, as shown for rapeseed protein-oleosome mixtures [94]. Oleosomes are oil droplets stabilised with phospholipids and membrane proteins, and upon adsorption at the interface, the droplets can rupture, spreading the oil, phospholipids and membrane proteins. In a protein-oleosome mixture, the oleosome rupture at the interface can significantly reduce the formation of stiff protein-stabilised interfaces [94]. If protein concentrations are high enough, the proteins can outcompete the adsorption of oleosomes at the air-water interface, thus preserving the protein-stabilised interface. In both the whey protein-escin and rapeseed protein-oleosome mixtures, the combination of imaging and rheology was essential to obtain the earlier-mentioned insights.

The GSD has also revealed specific insights on mixed systems. For example, in oleosome membrane proteins and phospholipid mixtures, the proteins formed stiff viscoelastic interfaces, and the phospholipids formed weak stretchable interfaces (J. [96]). Upon mixing, nearly identical Lissajous plots were observed for the mixtures at several ratios, suggesting that oleosins dominated the interface. But, after performing the GSD, a clear contribution of the phospholipids was observed, suggesting the presence of both components at the interface. A final example is an oleosome membrane-curcumin mixture-stabilised oil-water interface [97], where an increase of surface elastic modulus was observed at increasing deformation amplitude from 5 to 30 %, followed by a modulus decrease when increasing to 50 % deformation. This increase in modulus is unexpected, as higher deformation typically disrupts the interfacial microstructure, resulting in lower modulus values. Here, the GSD showed a continued increase of network interactions over the whole deformation range, while the surface density changes showed a similar increase followed by a decrease. These findings suggest that the interfacial network strength is increasing, likely due to newly adsorbed material being constantly compressed into a denser layer. At the same time, the density contribution increased, but decreased again at deformations >30 %. Ultimately, at >30 % deformations, the interfacial structure was broken down, resulting in an overall decrease in moduli. In both cases, the GSD was essential in gaining insights into the interface stabilising role of individual components in a multicomponent system.

4.4. Particles

Particles have been widely studied as interface stabilisers, often referred to as Pickering stabilisers, and have been studied using LAOD. Here, a main caution is the actual presence of particles at the interface in a diffusion-based system, as shown for protein-silica bead systems, where the beads sediment and change the droplet's shape, leading to an apparent surface tension decrease [116]. Sometimes, particles cannot spontaneously adsorb at an interface and attach only by convection, for instance, by using homogenisation. Several studies have demonstrated the absence of particles at the interface in tensiometry, as observed for soy protein fibrils [99]. These large fibrils were not surface-active, and the non-aggregated proteins in the solution dominated the interfacial

layer. The same was found for whey protein nanoparticles, where AFM imaging showed the absence of the particles on Langmuir-Blodgett films [24].

If the particles are actually adsorbing at the interface, LAOD can give insights into the type of layers formed. For example, for cellulose nanocrystals stabilised interfaces, salt screened the particle's charges and thus increasing the surface load and interactions at the interface [98]. Additional insights into the structural arrangement at the interface were obtained using neutron reflectometry. The GSD was used to study a gliadin nanoparticle system [23]. LAOD deformations were performed at different ageing times of the gliadin nanoparticle-stabilised interface. In combination with imaging, the authors showed how the nanoparticles transformed over time at the interface by slowly deforming and merging into a viscoelastic solid polymeric film.

5. Summary and future outlook

Large amplitude oscillatory dilatational (LAOD) rheology has emerged as a powerful technique for characterising the mechanical behaviour of interfaces under nonlinear deformation conditions. LAOD enables us to probe the complex viscoelastic properties of interfacial films stabilised by a broad range of surface-active molecules, including low molecular weight surfactants, proteins, particles, and their mixtures. These insights are crucial for understanding and predicting the stability of emulsions, foams, and other multiphase systems. For decades, the analysis of LAOD data was limited to first-harmonic-based moduli and Lissajous plots. While these methods provided valuable qualitative insights, they struggled to accurately quantify nonlinear behaviour, especially when higher harmonic contributions became significant. As such, first harmonic-based moduli in the nonlinear viscoelastic (NLVE) regime often led to incomplete or misleading interpretations of interfacial mechanics. The recent development of the general stress decomposition (GSD) has been a significant breakthrough. This new method enables a quantitative separation of rheological (network) contributions and surface density-driven effects in the stress response. By isolating the true rheological signal, GSD opens the door to a much more detailed and mechanistic understanding of interfacial behaviour. It allows rheological analyses that were previously not possible, thereby transforming the field of LAOD.

As the field advances, several promising research directions emerge:

- **Revisiting classic systems:** Many interfacial films previously characterised with traditional methods should be re-evaluated using LAOD in combination with GSD. This could reveal overlooked nonlinearities or misinterpreted behaviours.
- **Expanding GSD applications:** More interfacial systems, especially complex and multicomponent interfaces, should be studied using GSD to create a broader understanding of interfacial mechanics across different materials.
- **New rheological models**: Since GSD effectively decouples surface density changes from intrinsic rheology, the pure rheological signal obtained can serve as a foundation for new constitutive models of interfacial rheology. An example would be models to describe anisotropic interfacial behaviour.
- **Dynamic surface pressure isotherms**: The density-driven contribution identified using GSD directly links to the dynamic surface pressure behaviour. Systematically exploring this relationship could provide a better understanding of the origin of the formation of specific microstructures, such as those observed in protein-stabilised interfaces.

Together, these advances show that LAOD, especially in combination with GSD, gives a comprehensive framework for measuring and understanding nonlinear interfacial mechanics.

CRediT authorship contribution statement

Kerstin Risse: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Jack Yang:** Writing – original draft, Visualization, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Anteun de Groot:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Gerard Giménez-Ribes:** Writing – original draft, Methodology, Investigation. **Penghui Shen:** Writing – review & editing. **Stephan Drusch:** Writing – review & editing. **Emma B.A. Hinderink:** Writing – review & editing. **Leonard M.C. Sagis:** Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors have declared that no competing interest exist. This manuscript has not been published and is not under consideration for publication in any other journal. All authors approve this manuscript and its submission to Advances in Colloids and Interface Science.

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Data availability

Data will be made available on request.

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